MH3600 Topology and Manifolds

Tutorial #2

Topics covered:

- Topological spaces
- Bases and subbases
- The subspace topology

Problem 1

Let X be a topological space; let A be a subset of X. Suppose that for each $x \in A$ there is an open set U containing x such that $U \subset A$. Show that A is open in X.

Problem 2

(i) Let X be a set and let $\mathcal{T}_\alpha$ be the set of all subsets U of X such that $X - U$ either is countable or it is all of X. Show that $\mathcal{T}_\alpha$ is a topology on X.

(ii) Is the collection $\mathcal{T}_\infty = \{U \mid X - U \text{ is infinite or empty or all of } X\}$ a topology on X?

Problem 3

(i) If $\{\mathcal{T}_\alpha\}$ is a family of topologies on X, show that $\bigcap \mathcal{T}_\alpha$ is a topology on X. Is $\bigcup \mathcal{T}_\alpha$ a topology on X?

(ii) Let $\{\mathcal{T}_\alpha\}$ be family of topologies on X. Show that there is a unique smallest topology on X containing all the collections $\mathcal{T}_\alpha$, and a unique largest topology contained in all $\mathcal{T}_\alpha$.

(iii) If $X = \{a, b, c\}$, let

$$\mathcal{T}_1 = \{\varnothing, X, \{a\}, \{a, b\}\} \quad \text{and} \quad \mathcal{T}_2 = \{\varnothing, X, \{a\}, \{b, c\}\}.$$

Find the smallest topology containing $\mathcal{T}_1$ and $\mathcal{T}_2$, and the largest topology contained in $\mathcal{T}_1$ and $\mathcal{T}_2$.

Problem 4

Show that if $\mathcal{A}$ is a basis for a topology on X, then the topology generated by $\mathcal{A}$ equals the intersection of all topologies on X that contain $\mathcal{A}$. Prove the same if $\mathcal{A}$ is a subbasis.

Problem 5

Apply Lemma 13.2 in the textbook to show that the countable collection

$$\mathcal{B} = \{(a, b) \mid a < b, a \text{ and } b \text{ rational } \}$$

is a basis that generates the standard topology on $\mathbb{R}$.
Hint: use the "order density" of the rationals in the reals (see Exercise 4.9 part (d) in the textbook).

Problem 6

Show that if Y is a subspace of X and A is a subspace of Y, then the topology A inherits as a subspace of Y is the same as the topology it inherits as a subspace of X.